UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT
ON
Smart NoteVerse

Program Name: BCA

JAVA PROGRAMMING LAB

23CAP-352

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1. Abstract

The Simple Notepad Application is a desktop-based text editor developed using Java Swing. The primary objective of this project is to create a user-friendly application that allows users to perform basic text editing operations such as creating, opening, editing, and saving text files. The system also provides essential editing features like cut, copy, paste, and formatting options including font and color customization. This project helps in understanding the concepts of graphical user interface (GUI) design and file handling in Java. It serves as a foundational step toward building more advanced desktop applications.

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2. Introduction

In the modern computing environment, text editors play a crucial role in creating and modifying textual data. Almost every operating system includes a default text editor, such as Notepad in Windows. This project focuses on designing and implementing a similar lightweight text editor using Java.

Java provides powerful libraries like Swing that make it easier to build interactive GUI applications. Through this project, we aim to explore how Java can be used to create a fully functional desktop application.

The application is designed with simplicity in mind, ensuring that even beginners can use it without difficulty. It demonstrates how user inputs are handled, processed, and stored efficiently.

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3. Problem Statement

Many existing text editors are either too basic or too complex, making them unsuitable for beginners who need a simple tool for editing text files. There is a need for a lightweight application that provides essential features without overwhelming the user.

The problem addressed in this project is to design and develop a simple Notepad application that:

- Provides basic text editing features
- Is easy to use and understand
- Does not require complex setup or dependencies

- Efficiently handles file operations

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4. Objectives

The main objectives of this project are to develop a functional and efficient text editor while understanding core Java concepts.

Key objectives include:

- To design a graphical user interface using Java Swing
- To implement file operations such as open, save, and create new files
- To provide text editing functionalities like:
  - Cut
  - Copy
  - Paste
  - Select All
- To enable text formatting options like font and color
- To gain practical knowledge of event handling in Java

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5. System Overview

The system is a standalone desktop application that runs on any system with Java installed. It consists of multiple components working together to provide a seamless user experience.

The main components of the system include:

- **JFrame:** Acts as the main window of the application
- **JTextArea:** Used for writing and editing text
- **JMenuBar:** Provides access to different functionalities
- **JFileChooser:** Helps in selecting files for opening and saving

The system follows a simple workflow:

1. User interacts with the interface
2. The application processes the input
3. The result is displayed or saved accordingly

6. Methodology

The development of this project follows a structured approach to ensure proper implementation and testing.

Initially, the graphical interface is designed using Java Swing components. After designing the interface, menu options are added to provide different functionalities. Event handling is then implemented using the ActionListener interface to respond to user actions.

The methodology can be summarized as follows:

- Design GUI layout
- Add menu options and components
- Implement event handling
- Integrate file handling using Java I/O classes
- Test the application for different scenarios

This step-by-step approach ensures that each part of the application is properly developed and tested.

7. System Design

The system design describes how different components interact with each other. The application follows an event-driven design where user actions trigger specific operations.

Input

- User enters text using the keyboard
- User selects menu options

Processing

- The application processes commands like open, save, edit, and format

Output

- Text is displayed on the screen
- Files are saved or opened from storage

Modules

The system is divided into the following modules:

- **File Module**
  - New
  - Open
  - Save
  - Exit
- **Edit Module**
  - Cut
  - Copy
  - Paste
  - Select All
- **Format Module**
  - Change Font
  - Change Color

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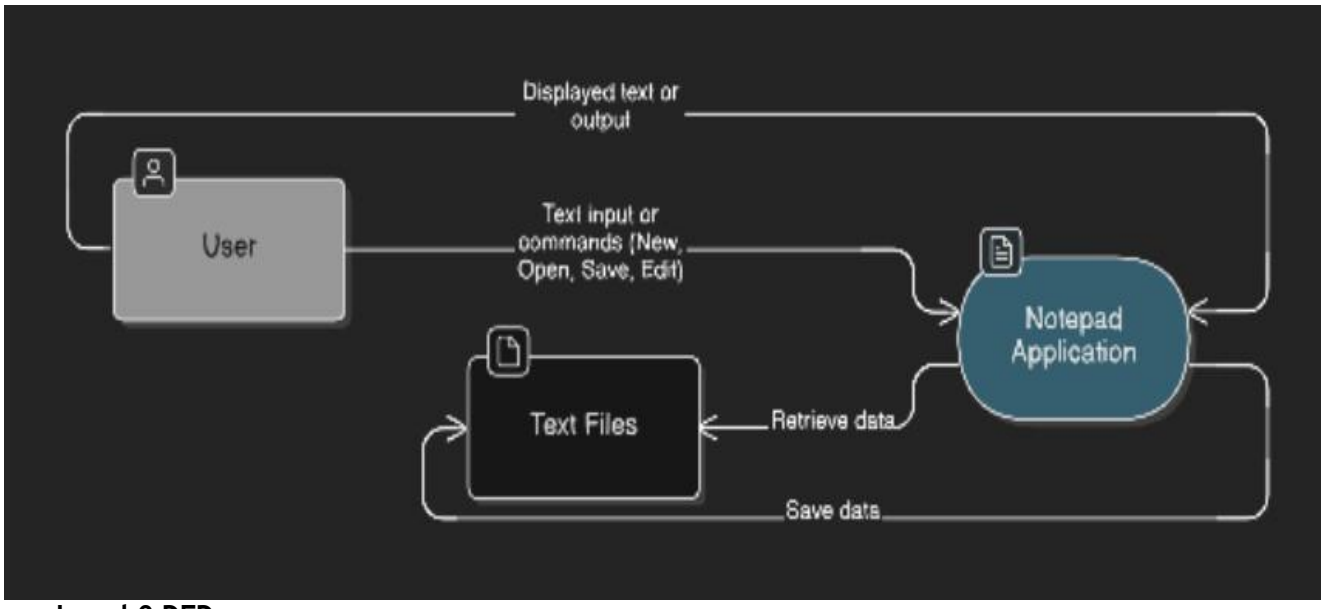
8. Data Flow Diagram (DFD)

The Data Flow Diagram represents how data moves within the system.



Level 0 DFD

- User → Notepad Application → File System

Explanation

- The user inputs text into the application
- The application processes user commands
- Data is either displayed or stored in files

G. Implementation

The implementation of the Notepad application is done using Java programming language and Swing library.

The application includes:

- A main class extending JFrame
- A JTextArea for text editing
- Menu components for different operations
- File handling using:
  - BufferedReader (for reading files)
  - BufferedWriter (for writing files)

Event handling is implemented using the ActionListener interface, which allows the program to respond to user actions like button clicks and menu selections.

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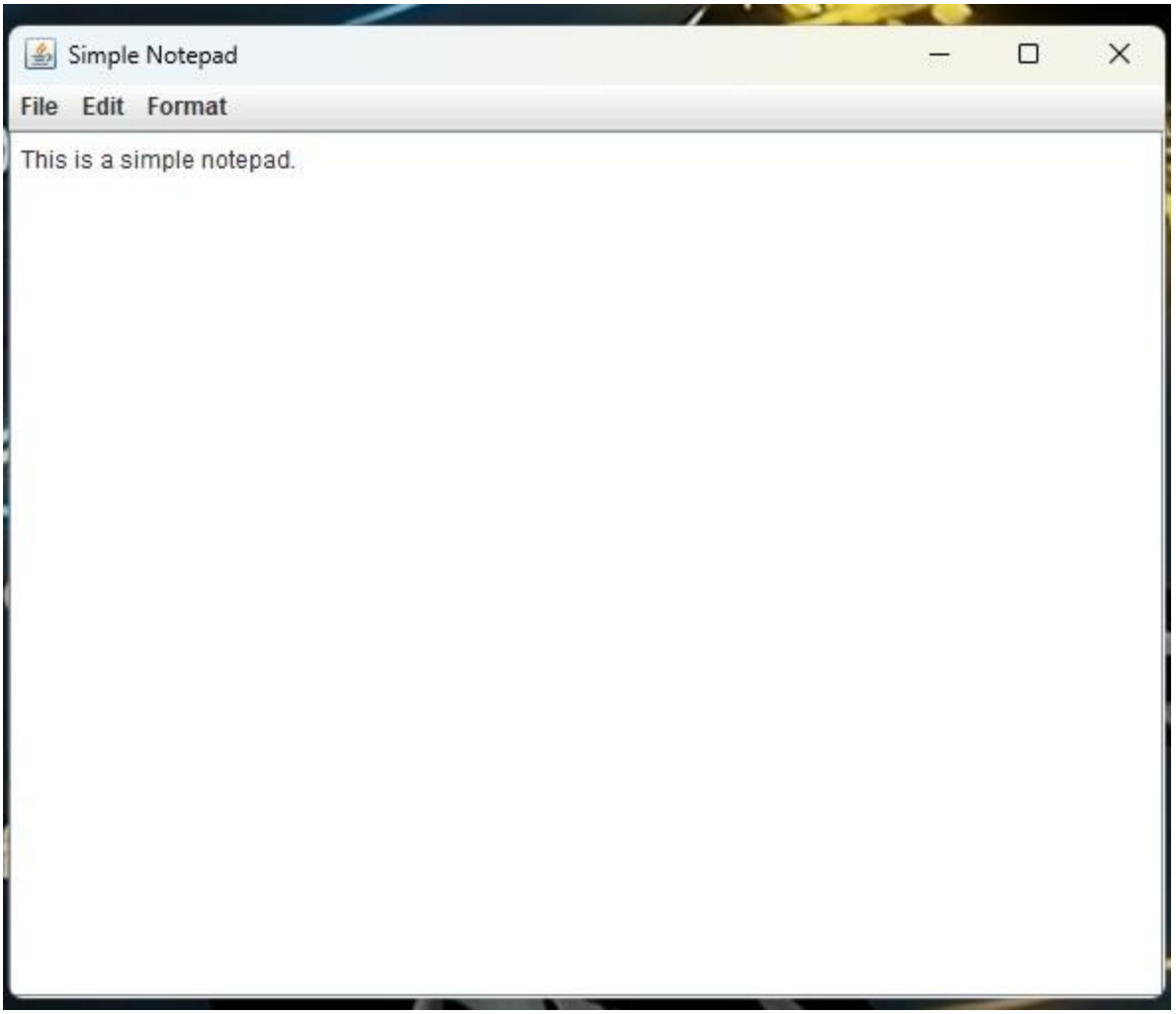
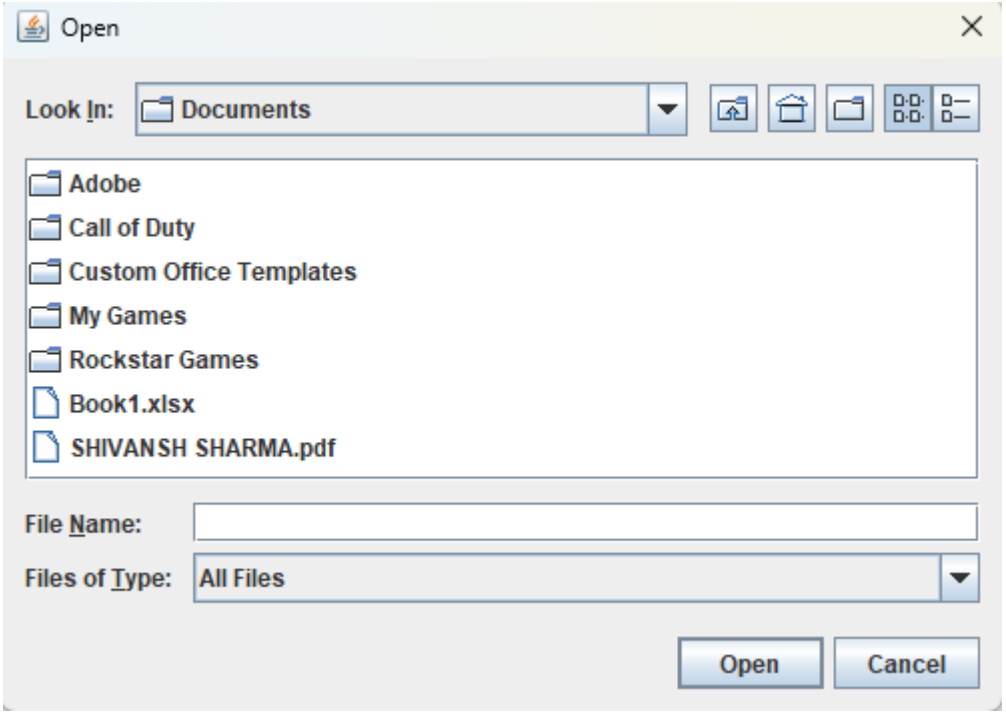
10. Results and Discussion

The application was successfully developed and tested. It performs all the intended functions effectively. Users can easily create, edit, save, and open text files without any difficulty.

The interface is simple and intuitive, making it suitable for beginners. The project demonstrates how Java can be used to build practical desktop applications.


Simple Notepad
File Edit Format
This is a simple notepad.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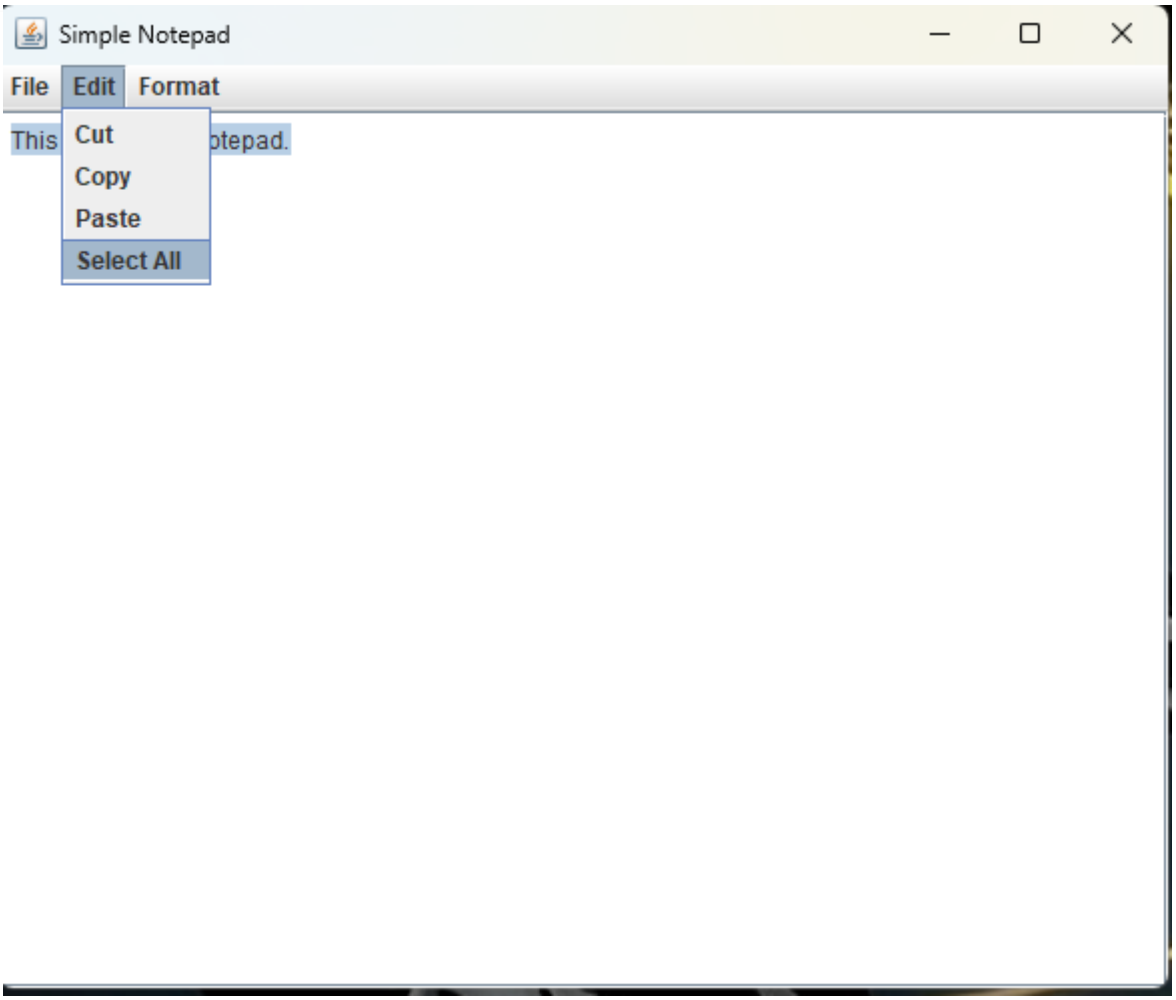

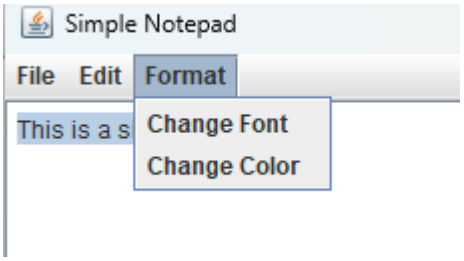

Observed results:

- Smooth file operations
- Responsive user interface
- Accurate text editing functions

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11. Advantages

The developed Notepad application provides several benefits in terms of usability, performance, and learning outcomes. Being a lightweight desktop application, it is designed to perform efficiently without consuming significant system resources. The

simplicity of the interface ensures that users, especially beginners, can easily interact with the system without requiring prior technical knowledge.

Another major advantage of this application is that it does not rely on any external database or complex backend system. Instead, it uses simple file handling mechanisms, making it easier to understand and implement. This makes the project highly suitable for educational purposes, particularly for students learning Java programming and GUI development.

Furthermore, the application supports essential text editing and formatting features, which are sufficient for basic document creation and editing tasks.

The key advantages of the system are as follows:

- The application provides a simple and user-friendly interface
- It is lightweight and executes efficiently on most systems
- No external database or server setup is required
- It helps in understanding Java Swing and event handling concepts
- It supports basic text editing and formatting functionalities

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12. Limitations

Despite its usefulness and simplicity, the Notepad application has certain limitations that restrict its functionality when compared to advanced text editors. The system is primarily designed for basic operations and does not include advanced features that are commonly found in modern editing tools.

One of the major limitations is the absence of advanced text processing features such as spell checking and grammar correction. Additionally, the application does not support working with multiple files simultaneously, which can reduce productivity for users who need to handle multiple documents at once.

Moreover, the formatting capabilities of the application are limited, as it only allows basic font and color changes without support for rich text formatting.

The main limitations of the system are as follows:

- The application supports only basic text editing features
- It does not include spell-check or grammar correction functionality
- Multiple document handling (multi-tab support) is not available
- Formatting options are limited compared to advanced editors

13. Future Scope

The project can be further enhanced by adding advanced features to improve functionality and user experience.

Possible improvements include:

- Find and Replace feature
- Word count display
- Multi-tab support
- Dark mode
- PDF export functionality

14. Conclusion

The Simple Notepad Application is a beginner-friendly project that demonstrates the use of Java Swing for GUI development and Java I/O for file handling. It successfully achieves its objective of providing a simple and efficient text editor.

This project helps in building a strong foundation in Java programming and prepares the developer for creating more advanced applications in the future.